Difference Enhancement and Interscale Interactive Fusion Mamba for Remote Sensing Image Change Detection

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***Abstract*—Recently, Mamba has made significant strides in sequence modeling, with its global receptive field, dynamic weighting strategy, and linear growth in computational complexity. In remote sensing (RS) change detection (CD), several studies have demonstrated that Mambas leverage a unique scanning mechanism to traverse images from various directions, showcasing excellent long-range modeling capabilities. However, as the network depth increases, Mamba often struggles to retain shallow textures and local features effectively. In particular, modern RS images frequently capture complex surface scenes, including seasonal climate variations and densely built environments, making local contextual details crucial for effective CD. Therefore, a difference enhancement and interscale interactive fusion Mamba (DEIF-Mamba) is proposed to alleviate the issue. This entire network framework integrates a convolutional neural network (CNN) and Mamba, utilizing CNN to capture local feature information, while Mamba employs a cross-scanning mechanism to integrate global information. To address the interference caused by mixed texture features and the missed detection of subtle changes in complex scenes, a differential feature enhancement module (DFEM) is proposed to enrich local contextual details and improve feature representation. In addition, we propose an interscale interactive fusion (ISIF) strategy to fully utilize the cross-scale interactive information and minimize information redundancy. Extensive experiments on four CD datasets demonstrate that the proposed DEIF-Mamba achieves an average F1 of 85.87% and shows superior performance compared with other state-of-the-art (SOTA) methods. Code will be available online at: https://github.com/Jyl199904/DEIF-Mamba**

***Index Terms*—Change detection (CD), Mamba, remote sensing (RS).**

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I. Introduction

REMOTE sensing (RS) change detection (CD) utilizes the same multitemporal land surface coverage area to identify surface changes, which provides the spatial distribution of features and is applied in various fields, encompassing urban expansion [1], deforestation [2], damage assessment [3], and forest cover mapping [4].

With the rapid advancement of satellites and drones, the acquisition of RS images with high temporal and spatial resolution offers robust support for CD. Nowadays, CD has traditional approaches [5] and deep learning (DL) approaches [6]. Traditional methods are straightforward in principle and highly operable; however, they do not fully leverage the feature information in RS images, often resulting in elevated false alarm rates in classification outcomes. Wu et al. [7] proposed a postclassification method that combines joint feature analysis and Bayesian fusion to reduce the false alarm rate by identifying the optimal probability of change categories. Liu et al. [8] integrated fuzzy C-means clustering and noise filtering to minimize errors during the image transformation, thereby reducing false detections. However, traditional approaches are struggling to meet the demands of complex feature representation in large-scale scenarios.

Recently, DL techniques have significantly advanced the field of CD, enabling the generation of more efficient and reliable change probability results. The mainstream models are mainly convolutional neural networks (CNNs) and Transformer. CNNs capture local features in RS images and perform deep fusion, effectively leveraging spatial context information. Zhan et al. [9] proposed a deep CNN for CD, trained with a contrastive weighted loss to extract more representative features. Daudt et al. [10] proposed a Siamese difference CNN that incorporates skip connections to capture rich encoding information in shallow layers, leading to more accurate predictions. To solve the problem of a series of tedious processes in traditional CNNs by generating depth feature difference maps and threshold segmenting changed pixels, Peng et al. [11] proposed an end-to-end UNet++ to extract multilevel fine-grained information and designed a fusion strategy to combine different semantic levels, which can generate high-precision change maps more efficiently. However, CNNs are limited by their local receptive field, which lead to insufficient

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understanding of global information. In 2017, Transformer [12] was proposed, which can model long-distance dependencies in the input sequence through the self-attention mechanism, thereby capturing the global feature information of the RS images. Yan et al. [13] grafted the pyramid structure of the progressive attention module to aggregate the multilayer features extracted from the Transformer to achieve enhanced effects and increase interdependence, solving the problem of irregularly changing areas in CD. To alleviate the inherent locality of convolution, Zhang et al. [14] designed a method to extract and fuse multiscale features using the Swin Transformer block as the basic unit and performed local calculations within each window through a sliding multihead self-attention mechanism, thus greatly alleviating the computational overhead problem. With advancements in CD, many researchers have discovered that the hybrid CNN–Transformer model effectively leverages the strengths of both architectures, leading to improved performance. To solve the interference caused by feature distribution differences due to spurious changes, Ji et al. [15] proposed a domain adaptive constraint loss for feature alignment and the interactive differential attention module to help the model better couple multiscale feature information based on the CNN–Transformer framework. To tackle the issue of low computational efficiency of the Transformer with long sequences, Mamba [16] was proposed, achieving linear scalability in sequence length and significantly enhancing inference efficiency. Subsequently, Mamba was successfully implemented in the visual field as well. Chen et al. [17] first combined the Mamba architecture to conduct spatiotemporal interaction of multitemporal features to obtain more accurate change information. Zhang et al. [18] proposed CDMamba, which integrates the scaled residual convMamba and adaptive guided fusion module to generate more discriminative difference features. Many studies have shown that Mambas achieve effective detection by efficiently capturing global information through their unique scanning mechanism. However, in complex scenes with seasonal climate changes or densely distributed buildings, interference from mixed texture features such as snow cover or building shadows is prone to occur. In addition, as the number of network layers increases, Mamba tends to overlook important local textures and detailed information, which lead to certain limitations in detecting scale or subtle changes.

To effectively address these challenges, we propose a difference enhancement and interscale interactive fusion Mamba (DEIF-Mamba). First, DEIF-Mamba extracts the feature information of RS images within each layer through the CNN backbone. Then, the dual-temporal (DT) features from the last layer are fed into the designed differential feature enhancement module (DFEM) to better capture local feature details, thereby achieving a richer feature representation. Next, we sequentially feed the multiscale DT features into the Mamba-structured decoding layer to capture spatiotemporal associations. To better distinguish dynamic changes from the static background, we pass the features through the interscale interactive fusion (ISIF) to decompose and interactively fuse the features at different scales, strengthen the use of temporal information, and highlight the change characteristics. Finally, we output a binary change result through a 1×1 convolutional layer. The contributions of our work can be summarized as follows.

1) We design a hybrid CNN-Mamba network that employs CNNs to extract local features and Mamba to model global spatiotemporal information, fully leveraging the strengths of both to enhance detection performance.
2) The DFEM is proposed to enhance the capture of local contextual detail information and improve feature representation, thereby alleviating the interference of mixed texture features and the missed detection of subtle changes in dense building scenes.
3) The ISIF module is proposed to improve the model's ability to effectively utilize cross-scale interactive temporal information from DT images while minimizing information redundancy.

The rest of this article is organized as follows. Section II describes the related work. Section III gives the details of our proposed method. Experimental results are reported in Section IV. The conclusion is made in Section V.

II. Related Works

A. CNN-Based Method

In conventional CD, researchers primarily depended on manual feature extraction, which often suffers from instability and limited generalizability. However, with the advent of DL, CNNs have demonstrated exceptional capability in capturing high-level image features, progressively superseding traditional approaches. Zhan et al. [9] proposed a deep Siamese convolutional network that directly extracts more abstract and robust features on a pixel-by-pixel basis and then uses feature distance measurement to quantify the changes between image pairs. He et al. [19] proposed the residual network and introduced identity connections, which enable direct information flow between different layers of the network, effectively alleviating the problem of network degradation. For CD, capturing complex surface features at a single scale presents certain limitations. For instance, the changing characteristics of water bodies, forests, and urban buildings vary across different spatial scales. Therefore, multiscale processing is often employed to incorporate richer texture and shape feature information. Zhang and Shi [20] developed a multiscale feature difference network and integrated it with a training strategy that leverages a small number of pixel-level samples, enhancing the recognition of changing features. Wu et al. [21] proposed a multiscale graph convolutional network that distributes label information across nodes through progressive relationships between scales, ultimately employing a fusion strategy to integrate multiscale feature information. Most studies have shown that incorporating attention mechanisms can help the model adaptively weigh feature importance, thereby enhancing feature representation. Lv et al. [22] designed a spatial–spectral attention mechanism that leverages rich contextual information to guide the model's focus on land cover change areas and reduce noise interference. Shi et al. [23] proposed a deep supervised metric network that combines a convolutional block attention module (CBAM) to enhance discriminative change features and improve detection capabilities. However, traditional CNNs

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are limited to capturing relationships between adjacent pixels and struggle to effectively model distant dependencies. With the continued improvement in the quality of images acquired by sensors, the inherent local receptive field of CNNs alone struggles to effectively handle complex ground object backgrounds, making them insufficient for efficiently addressing CD tasks.

B. Transformer-Based Method

Transformer [12], model global dependencies by self-attention mechanism, contributing significantly to vision tasks such as super-resolution [24], [25], classification [26], [27], and segmentation [28], [29]. In CD, Transformer can not only effectively process high-dimensional data structures but also capture global feature information through multihead self-attention, thereby enhancing feature representation. Chen et al. [30] introduced a Transformer-based CD network designed to achieve high-precision change results, which leverages the encoder to model spatiotemporal context information and employs the decoder to refine the original feature representations. Wang et al. [31] employed the Transformer architecture integrated with CBAM to extract multiscale features, effectively modeling spatial and channel information. Numerous experiments have demonstrated that the CNN–Transformer hybrid model effectively integrates local feature extraction with global modeling capabilities. By fully leveraging the strengths of both architectures, this approach significantly enhances the robustness of CD [32]. Wang et al. [33] proposed MCTNet, which consists of multiple modules, including CNN and Transformer, efficiently extracting key features from complex time-series data. To alleviate the lack of sensitivity of existing hybrid models in scale-changing regions, Li et al. [34] proposed the ConvTrans block, which extracts multiscale global–local features and then adaptively fuses multilayer features to enhance the saliency of changing regions. However, the CNN-Transformer often produces redundant information due to the local nature of convolution operations, leading to an overabundance of similar features from neighboring pixel regions. To efficiently model the relationship between feature layers and fully capture global–local context information, Li et al. [35] proposed TransUNetCD, which combines the strengths of the Transformer and UNet architectures, enables the extraction of rich change information, and allows for pixelwise weighting to select the aggregation of features. Jiang et al. [36] proposed MSFCTNet, which filters redundant feature information layer by layer using a gated attention module, and then performs feature refinement to effectively suppress background noise. Although Transformer offers significant advantages in CD, its computational complexity increases sharply with the number of DT image slices due to the self-attention mechanism. In addition, the relationship matrix between long sequences requires substantial memory, further exacerbating the computational burden.

C. Mamba-Based Method

Recently, the sequence model named Mamba [16] has demonstrated the ability to efficiently model long sequences by incorporating the selective state-space mechanism. To a large extent, Mamba alleviates the challenges related to computational complexity and memory consumption that arise from traditional self-attention mechanisms. In the field of computer vision (CV), Zhu et al. [37] proposed bidirectional Mamba blocks (Vim) that utilize position embedding to label image sequences and compress visual representations. Numerous experiments have shown that Vim can perform more efficient performance across tasks, such as classification, object detection, and semantic segmentation, simultaneously. Chen et al. [17] introduced Vim in CD for the first time, leveraging its ability to fully model global context information and facilitate spatiotemporal interactions, which outperforms traditional CNN and Transformer models in terms of performance. However, the existing Mamba model obtains the global receptive field through the selective scanning mechanism, which neglects the importance of local information in CD. Zhang et al. [18] proposed scaling residual ConvMamba that captures detailed feature information, thereby enhancing the ability to discriminate changing features. Liu et al. [38] proposed a tree-topology Mamba-guided network (TTMGNet), which effectively extracts global features and incorporates spatial and positional information. TTMGNet preserves fine local details and suppresses background noise, improving the accuracy of CD. Although Mamba can effectively capture long-range dependencies, it lacks explicit consideration of spatially varying content, resulting in suboptimal pixel-wise representation during local modeling [39]. Therefore, although many Mambas demonstrate efficient performance in CD, they tend to ignore local texture detail information in complex and dense scenes, which leads to certain limitations in detecting scale changes or subtle changes. In this article, we propose DEIF-Mamba, a model designed to capture local detail information while integrating global spatiotemporal context information to achieve fine-grained detection.

III. Methodology

A. Preliminaries

As the core architecture of Mamba, state-space models (SSMs) have gained prominence in DL for the ability to map sequence data to state spaces, effectively capturing long-term dependencies. SSM represents the current state by defining a set of equations that describe the relationship between inputs $\mathrm{P}x(t) \in \Re^p$ and outputs $\mathrm{P}y(t) \in \Re^q$

$$h'(t) = Ah(t) + Bx(t) \tag{1}$$

$$y(t) = Ch(t) \tag{2}$$

where $\mathrm{P}A \in \Re^{n\times n}$ and $\mathrm{P}B \in \Re^{n\times 1}$, respectively, represent the state transition matrix and input matrix, $\mathrm{P}C \in \Re^{1\times n}$ represents the observation matrix, $h(t)$ represents the state at time t, $h'(t)$ represents the derivative of the state vector $x(t)$ with respect to time t.

The subsequent is discretized through a zero-order hold (ZOH) operation, which includes a time scale parameter Δ to turn the continuous

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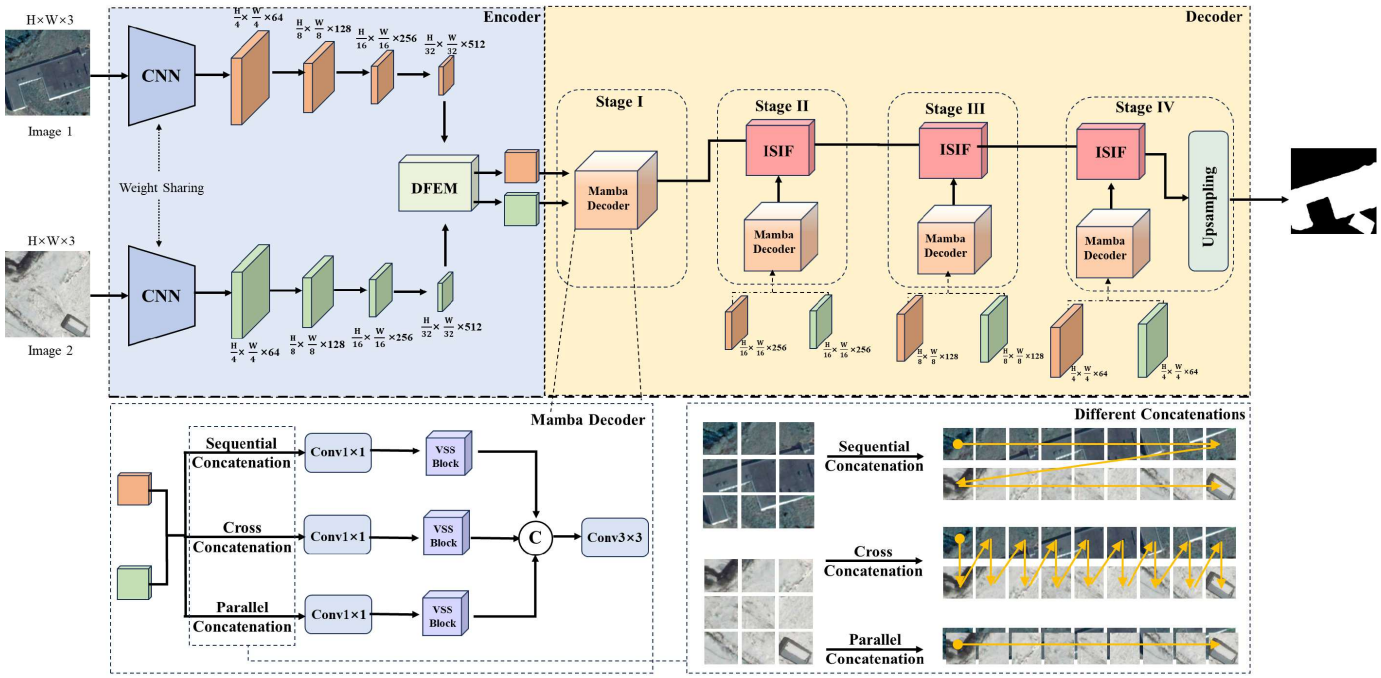


Fig. 1. Architecture of the proposed DEIF-Mamba.

parameters A and B into the discrete parameters $\bar{A}$ and $\bar{B}$

$$\bar{A} = \exp(\Delta A) \tag{3}$$

$$\bar{B} = (\Delta A)^{-1}(\exp(\Delta A) - I) \cdot \Delta B$$

$$h(t_{k+1}) = \bar{A}h(t_k) + \bar{B}x(t) \tag{4}$$

$$y(t) = Ch(t). \tag{5}$$

Finally, the final output is obtained through a convolution operation

$$\bar{K} = (C\bar{B}, C\overline{AB}, \ldots, C\overline{A^{N-1}B}) \tag{6}$$

$$y(t) = x(t) * \bar{K} \tag{7}$$

where N is the length of the input sequence and $\bar{K}$ is the structured convolution kernel.

To address the Mamba's limitation of effectively modeling visual data sequences, Visual Mamba [37] introduces the visual state-space (VSS) block based on SSM, which traverses the images through cross scanning and obtains the global receptive field. In the proposed method, the VSS block will serve as the core component of the entire network.

B. Architecture Overview

The proposed DEIF-Mamba architecture, as shown in Fig. 1, processes a DT image pair through a shared-weight CNN encoder to extract multiscale DT feature pairs. The deepest DT feature pair is refined by the DFEM to capture local context details, and these refined features are then decoded through a four-stage decoder. At each decoding stage, the spatiotemporal relationships of the DT feature pair are modeled using a Mamba Decoder with the same structure. Stage I initializes decoding by applying a Mamba Decoder to the DFEM-refined DT feature pair. For Stages II–IV, the DT feature pair at each corresponding scale is first refined by a Mamba Decoder. Subsequently, the Mamba-refined DT features are combined with the output from the preceding stage, and this dual-source representation is passed to the ISIF module to interactively decompose and fully exploit spatial context information. Finally, a classifier is applied to the final decoded representation to produce the binary change map.

C. CNN Encoder

DEIF-Mamba employs the pretrained ResNet-18 [19] as the CNN encoder to extract both shallow and deep features from RS images, due to its superior feature extraction capability and effective preservation of fine-grained spatial details across low-level scales [40]. The entire backbone primarily comprises a 7×7 convolutional layer and four residual blocks. Initially, the DT images are transformed into the low-level feature representation, with their resolution reduced via the first convolutional layer. The four residual blocks sequentially capture edges, local structures, and abstract semantic information through two 3×3 convolutional layers. Second, residual connections are established between each residual block, enabling the network to retain shallow features and integrate them with deep features, thereby mitigating the gradient vanishing problem. DT features at four different scales will be selected for subsequent spatiotemporal context information modeling, as shown in the following formula:

$$F_1', F_1'', F_1''', F_1^{\mathrm{iv}} = \mathrm{Res}(\mathrm{Img1})$$

$$F_2', F_2'', F_2''', F_2^{\mathrm{iv}} = \mathrm{Res}(\mathrm{Img2}) \tag{8}$$

where Res represents the pretrained ResNet-18.

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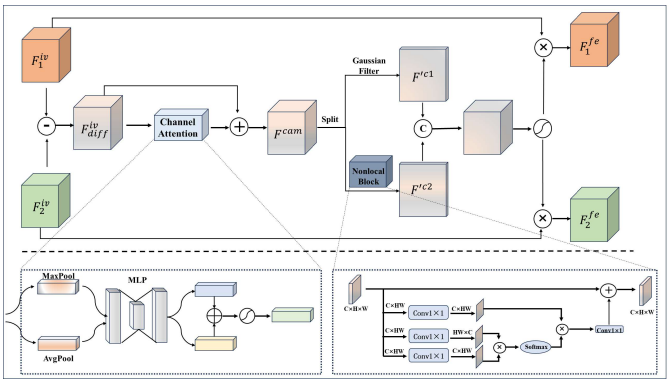


Fig. 2. Architecture of the proposed DFEM.

D. Differential Feature Enhancement Module

Nowadays, complex RS scenes contain various interference factors that challenge CD tasks. For instance, vegetation or bare land may experience significant seasonal variations in time series, leading to false detections of real changes and background differences. Densely packed buildings often exhibit highly similar appearances, which may cause small-scale variations to be overlooked. Local changes within the same building, such as renovations or color alterations, may be difficult to distinguish from the surrounding area, causing these local feature changes to be masked. To address this, we propose the DFEM to better identify subtle surface or building changes and differentiate pseudo-change interference by enhancing the model's ability to capture local context information.

First, we compute the absolute difference of the last layer of the DT feature and apply channel attention (CA) [41] to the difference features as depicted in Fig. 2

$$F_{\text{diff}}^{\text{iv}} = |F_1^{\text{iv}} - F_2^{\text{iv}}| \tag{9}$$

$$F^{\text{cam}} = F_{\text{diff}}^{\text{iv}} + M_c * F_{\text{diff}}^{\text{iv}}$$
$$M_c = \sigma\left(\text{MLP}\left(\text{Avgpool}\left(F_{\text{diff}}^{\text{iv}}\right)\right) + \text{MLP}\left(\text{Maxpool}\left(F_{\text{diff}}^{\text{iv}}\right)\right)\right) \tag{10}$$

where σ represents the sigmoid function, M_c represents the attention weight, F^{cam} represents the feature after CA.

We use CA to reduce the weights of redundant channels in the features and enhance the response to key feature information. Subsequently, the CA enhanced features are split into two parts with equal proportion along the channel dimension. One part is smoothed using Gaussian filtering with a specified Gaussian kernel, while the other part undergoes nonlocal fast computation to determine the correlations between all pixels in the feature map. We set the Gaussian kernel size to 3×3, the bandwidth parameter to 1.0, and construct a multistage adaptive feature optimization module in the nonlocal block architecture, implementing five stacked layers of attention-based feature optimization units. Each pixel is reassigned based on the generated attention weights, resulting in features that capture long-range dependencies

$$F^{c1}, F^{c2} = \text{split}\left(F^{\text{cam}}\right) \tag{11}$$

$$F'^{c1} = \sum_{m=[k/2]}^{[k/2]} \sum_{n=[k/2]}^{[k/2]} G(m,n) \cdot F^{c1}(i+m, j+n) \tag{12}$$

$$F'^{c2} = F^{c2} + W_z \cdot \frac{1}{d} \sum_{i,j=1}^{N} f(F^{c2}(i), F^{c2}(j)) g(F^{c2}(j)) \tag{13}$$

where k represent the Gaussian kernel size, $G(\cdot,\cdot)$ represents the Gaussian kernel weight. W_z and d are the projection matrix and the normalization factor, $f(\cdot,\cdot)$ and $g(\cdot)$ are the similarity function and feature change, respectively.

Through the above design, which not only enables the model to retain structural information in the spatial domain and effectively suppresses background noise but also captures long-range dependencies and enhances global feature representation. We then merge the two features and pass them through a sigmoid function to obtain the probability distribution. Finally, the result is multiplied by the input feature to produce the enhanced output

$$F_1^{\text{fe}} = F_1^{\text{iv}} * \sigma\left(\text{concat}\left(F'^{c1}, F'^{c2}\right)\right)$$
$$F_2^{\text{fe}} = F_2^{\text{iv}} * \sigma\left(\text{concat}\left(F'^{c1}, F'^{c2}\right)\right). \tag{14}$$

When faced with the mixed texture feature interference of complex surface scenes and the subtle local variation features in dense building environments, the DFEM effectively enhances low-frequency features and improves the model's ability to capture context-detailed information. In addition, integrating local detailed information with global spatiotemporal context information can more effectively identify change characteristics.

E. Mamba Decoder

The Mamba Decoder adopts the architecture of the spatiotemporal state-space (STSS) module proposed in MambaBCD [17], which is designed to effectively model spatiotemporal relationships in DT RS images. Its structure consists of three main components (see Fig. 1): spatiotemporal concatenations, VSS blocks, and a convolutional fusion layer. Specifically, the spatiotemporal concatenations include sequential concatenation, cross concatenation, and parallel concatenation, which organize features from pre-event and postevent images in different ways to explore temporal dependencies and intertemporal interactions. Each concatenated feature set is processed by a VSS block, which leverages the Mamba architecture to capture global spatial and temporal context. Finally, the outputs of the VSS blocks are fused through a channelwise concatenation operation followed by a convolutional layer to refine the learned features. This design enables the decoder to fully exploit spatiotemporal relationships, contributing to accurate and robust CD.

F. Interscale Interactive Fusion

The fusion of features at different scales enables the model to effectively integrate spectral and textural shape features, facilitating the differentiation of dynamic changes from the static background. However, most fusion methods rely on simple linear combinations, which fail to fully exploit spatial

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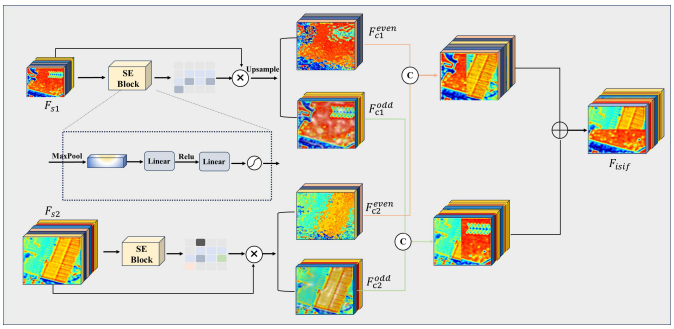


Fig. 3. Architecture of the proposed ISIF.

information and often lead to feature redundancy. To address this issue, we propose the ISIF that decomposes and interactively fuses features across different scales. The ISIF can enhance the utilization of spatial information and improve the model's sensitivity to surface change regions, thereby effectively highlighting change features.

As illustrated in Fig. 3, after the Mamba Decoders capture the global spatiotemporal context information, we fuse the features at different scales to effectively integrate spatial information and highlight the change areas. First, we input features at two different scales (F_{s1} and F_{s2}) into the squeeze-and-excitation (SE) module [42] to generate attention weights that respond to the changes. We then multiply the attention weights with the input features and upsample the small-scale features to obtain the extracted features F_{c1} and F_{c2}. To take the F_{c2} as an example, the calculation process can be described as

$$F_{c2} = F_{s2} * \sigma(W_2 \text{Relu}(W_1(\text{Maxpool}(F_{s2})))) \tag{15}$$

where W_1 and W_2 represent the weight matrices from the SE module.

Finally, we decompose the two extracted features into even and odd indices along the channel dimension and perform interactive fusion

$$F_{c1}^{\text{even}}, \quad F_{c1}^{\text{odd}} = \text{DC}(F_{c1}) \tag{16}$$

$$F_{c2}^{\text{even}}, \quad F_{c2}^{\text{odd}} = \text{DC}(F_{c2}) \tag{17}$$

$$F_{\text{isif}} = \text{concat}(F_{c1}^{\text{even}}, \quad F_{c2}^{\text{odd}}) + \text{concat}(F_{c2}^{\text{even}}, \quad F_{c1}^{\text{odd}}) \tag{18}$$

where DC refers to the decomposition function, which separates features into even and odd indices along the channel dimension.

Through the above fusion strategy, feature information complementarity at the fine-grained level is achieved, which effectively reduces information redundancy. Furthermore, the ISIF significantly enhances detection performance and substantially reduces computational complexity in complex RS images.

G. Prediction Head and Loss Function

In the final stage of change map generation, the number of channels is mapped to two using a 1×1 convolutional layer. Subsequently, the output is mapped to the same size with the original image via bilinear interpolation. During the training phase, we perform backpropagation to optimize the model parameters by approximating the cross-entropy loss L_{ce} between the prediction and the ground-truth toward convergence

$$L_{\text{ce}}(Y, P) = -\frac{1}{N} \sum_{i=1}^{N} \left[Y_i \log_2 P_i + (1 - Y_i) \log_2 (1 - P_i) \right] \tag{19}$$

where N represents the total number of pixels, Y_i is the true label of the pixel i, and P_i is the correct predicted probability of pixel i. In the inference phase, a single-channel CD image is output through the argmax operation.

IV. Experiments

A. Data Description

To evaluate the performance of the DEIF-Mamba on quantitative and qualitative metrics, we conducted extensive experiments on four representative RS CD datasets.

1) *CDD [43]:* The CDD dataset primarily consists of RS images capturing variations induced by actual seasonal climate. These images are sourced from Google Earth, with resolutions ranging from 0.03 to 1 m.
2) *LEVIR-CD [1]:* The learning, vision (LEVIR-CD) primarily comprises 637 RS images, each spanning a period of at least five years, with a resolution of 0.5 m. It encompasses substantial land use changes, including the growth and decline of buildings.
3) *Change in Land Cover Dataset (CLCD) [44]:* The CLCD primarily focuses on changes in farmland, increases in bare land, and forest degradation, with a resolution of approximately 2 m. The RS images are mainly sourced from Sentinel-2 and Landsat-8, covering representative regions in China, such as the Yangtze River Delta.
4) *WHU-CD [45]:* The WHU-CD mainly covers the earthquake reconstruction area, reflecting the changes in surface buildings and other land cover types. Its resolution is 0.5 m, derived from high-resolution satellite and aerial photography.

B. Experimental Setup

1) Comparison With State of the Arts (SOTAs): To assess the effectiveness and accuracy of DEIF-Mamba, we compared it to several SOTA approaches, which include IFNet [46], SNUNet [47], BiT [30], DASNet [45], STANet [1], MSCANet [32], DMINet [48], and MambaBCD [17].

IFNet [46] employs a fully convolutional two-stream architecture to extract deep features and then utilizes differential reconstruction to generate precise change maps.

SNUNet [47] leverages the dense connections of NestedUNet to extract detailed change features and introduces the CA to suppress positioning errors.

BiT [30] utilizes the Transformer-based encoding layer to model contextual information, and features are projected back to the pixel space and refined to enhance spatial details.

DASNet [45] employs dual attention to extract discriminative variation features and introduces the double-margin contrastive loss to mitigate the issue of category imbalance.

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TABLE I
COMPARISON RESULTS ON THE LEVIR-CD AND CDD DATASETS. THE OPTIMAL RESULTS ARE HIGHLIGHTED IN RED, AND THE SUBOPTIMAL RESULTS ARE HIGHLIGHTED IN BLUE. ALL RESULTS ARE EXPRESSED IN PERCENTAGE (%)

Methods	LEVIR-CD						CDD					
	Precision	Recall	F1	IOU	Kappa	OA	Precision	Recall	F1	IOU	Kappa	OA
IFNet	79.20	80.49	79.84	66.15	79.25	98.56	93.42	89.31	91.32	87.24	92.32	98.29
SNUNet	77.79	82.39	80.03	66.70	79.33	98.61	93.51	92.42	92.96	87.71	92.59	98.31
BiT	77.23	78.57	77.89	65.17	77.71	97.39	93.34	91.93	92.63	87.58	92.76	98.31
DASNet	79.66	76.28	77.93	65.36	77.87	97.90	93.57	90.91	92.22	87.40	92.48	98.06
STANet	80.92	74.33	77.48	65.06	77.16	97.69	93.68	90.60	92.11	87.27	92.09	98.22
MSCANet	79.14	80.36	79.74	67.35	79.55	98.64	94.87	95.16	95.02	89.51	93.35	98.81
DMINet	81.47	80.25	80.86	67.87	80.21	98.75	94.99	94.22	94.60	89.76	93.38	98.73
MambaBCD	77.57	83.47	80.41	67.24	79.22	98.67	95.51	93.80	94.65	89.84	93.94	98.75
DEIF-Mamba	78.71	84.14	81.33	68.54	80.68	98.74	95.65	95.68	95.67	91.70	95.89	98.98

STANet [1] captures dependencies of different scales based on Siamese spatiotemporal attention to adapt to the scale changes of objects.

MSCANet [32] extracts multiscale features through the CNN backbone and aggregates contextual information through the Transformer decoding layer.

DMINet [48] proposes cross-temporal joint attention to control the global feature distribution and stimulate information coupling.

MambaBCD [17] first attempt to use the Mamba architecture for CD, fully learning the global spatial context information and modeling the spatiotemporal relationship to obtain accurate change results.

2) Experimental Details: The proposed DEIF-Mamba is experimented on an NVIDIA RTX 4090D, using the PyTorch framework as the model development environment. During the training process, the Adam optimizer was employed for model optimization, with a learning rate of $1e^{-4}$. The training was conducted over 150 epochs with a batch size of 8.

3) Evaluation Metrics: To evaluate the performance and accuracy of DEIF-Mamba, we use six evaluation metrics to measure the agreement between the predicted result and the ground truth, namely, precision, recall, F1-score (F1), intersection over union (IOU), overall accuracy (OA), and Kappa. Each indicator is defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{20}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{21}$$

$$F1 = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{22}$$

$$\text{IOU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{23}$$

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{24}$$

$$\text{Kappa} = \frac{\text{OA}-P}{1-P} \tag{25}$$

$$P = \frac{(\text{TP}+\text{FP})(\text{TP}+\text{FN})+(\text{FN}+\text{TN})(\text{TP}+\text{TN})}{(\text{TP}+\text{FP}+\text{TN}+\text{FN})^2} \tag{26}$$

where TP, TN, FP, and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively.

C. Performance Comparison and Result Analysis

As shown by the quantitative indicators in Tables I and II, our proposed DEIF-Mamba has a better performance than other SOTAs. On LEVIR-CD, DEIF-Mamba improves F1 by 0.92% and kappa by 1.46% compared with the MambaBCD. However, compared with DMINet and MSCANet, the precision demonstrates a reduction ranging from 0.43% to 2.76%. We attribute this to the cross-temporal joint attention mechanism in DMINet and the multiscale context information aggregation block in MSCANet, which effectively enhance the model's ability to reduce the misjudgment rate in detecting changing areas. Across several other metrics, DEIF-Mamba demonstrates significant advantages. On the CLCD, DEIF-Mamba achieves improvements across five metrics. Specifically, compared with MambaBCD, the F1 improves by 2.95% and 4.27% in IOU. It is evident that DEIF-Mamba achieves a higher detection accuracy when addressing farmland changes induced by seasonal variations. On the CDD and WHU-CD, DEIF-Mamba compared with MambaBCD achieves improvements of 1.02% and 2.56% in F1 and 1.86% and 4.16% in IOU, respectively. These quantitative results demonstrate that the proposed DFEM and ISIF enable the model to achieve higher accuracy compared with other SOTAs.

To verify the sensitivity of the proposed module to subtle local variation features, especially in scenarios involving seasonal climate variations and densely packed building complexes, we conducted visualization experiments on four

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TABLE II

COMPARISON RESULTS ON THE CLCD-CD AND WHU-CD DATASETS. THE OPTIMAL RESULTS ARE HIGHLIGHTED IN RED, AND THE SUBOPTIMAL RESULTS ARE HIGHLIGHTED IN BLUE. ALL RESULTS ARE EXPRESSED IN PERCENTAGE (%)

Methods	CLCD-CD						WHU-CD					
	Precision	Recall	F1	IOU	Kappa	OA	Precision	Recall	F1	IOU	Kappa	OA
IFNet	68.92	65.54	67.18	52.36	66.89	95.49	85.56	86.63	86.09	76.46	86.09	98.85
SNUNet	66.77	62.41	64.52	47.62	61.78	94.89	73.91	83.39	78.36	64.43	77.42	98.17
BiT	72.49	71.06	71.77	55.96	69.52	95.84	86.78	88.52	87.64	78.01	87.12	99.01
DASNet	71.42	69.57	70.48	54.42	68.15	95.66	75.73	89.42	82.01	69.51	82.10	98.44
STANet	70.11	68.78	69.43	54.13	68.03	95.61	85.07	87.97	86.49	77.41	86.45	98.87
MSCANet	75.57	72.29	73.90	58.60	71.85	96.20	88.46	89.04	88.75	79.78	88.29	99.10
DMINet	74.31	72.72	73.50	58.11	71.40	96.10	90.29	88.75	89.51	81.02	89.08	99.17
MambaBCD	76.54	70.13	73.20	57.22	71.14	96.17	90.00	85.62	87.76	78.19	87.26	99.05
DEIF-Mamba	78.73	73.75	76.15	61.49	74.31	96.57	92.89	87.89	90.32	82.35	89.91	99.25

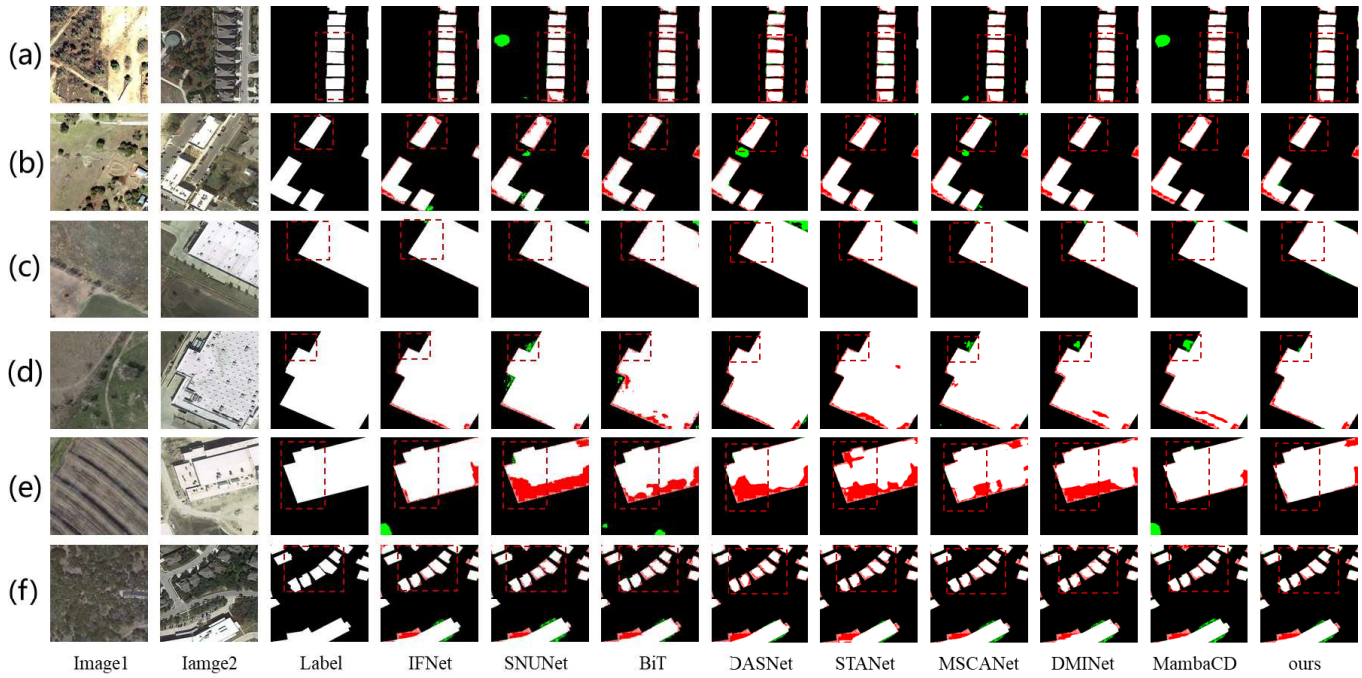


Fig. 4. Visual results of different models on test samples of LEVIR-CD. (a)–(f) Six representative samples.

datasets. Figs. 4–7 show the visualization results of each method across four datasets. We use different colors to indicate the detection effect: true positives (TPs, white), true negatives (TNs, black), false positives (FPs, red), and false negatives (FNs, green). In addition, we use the red dotted box to mark the target, so as to better highlight the excellent performance of the results we present.

Visualization on LEVIR-CD: Fig. 4(a), (b), and (f) illustrates the interwoven distribution of densely packed buildings and bare soil, accompanied by bushes that may act as interfering factors. Fig. 4(c), (d), and (e) reflects the construction of large buildings and road construction. The visualization results intuitively demonstrate that DEIF-Mamba exhibits superior performance in detecting areas with subtle local changes when dealing with densely packed buildings. In addition, it is apparent that ponds or bushes, which share similar color characteristics with buildings, may lead to misinterpretations. By integrating DFEM and ISIF, we can extract more detailed features and enhance feature representation, thereby mitigating such interference.

1) Visualization on CDD: The selected images illustrate changes driven by seasonal climate variations. Fig. 5(a)–(c) depicts the changes in houses during extreme cold weather, while Fig. 5(d)–(f) shows the changes in roads across differ-

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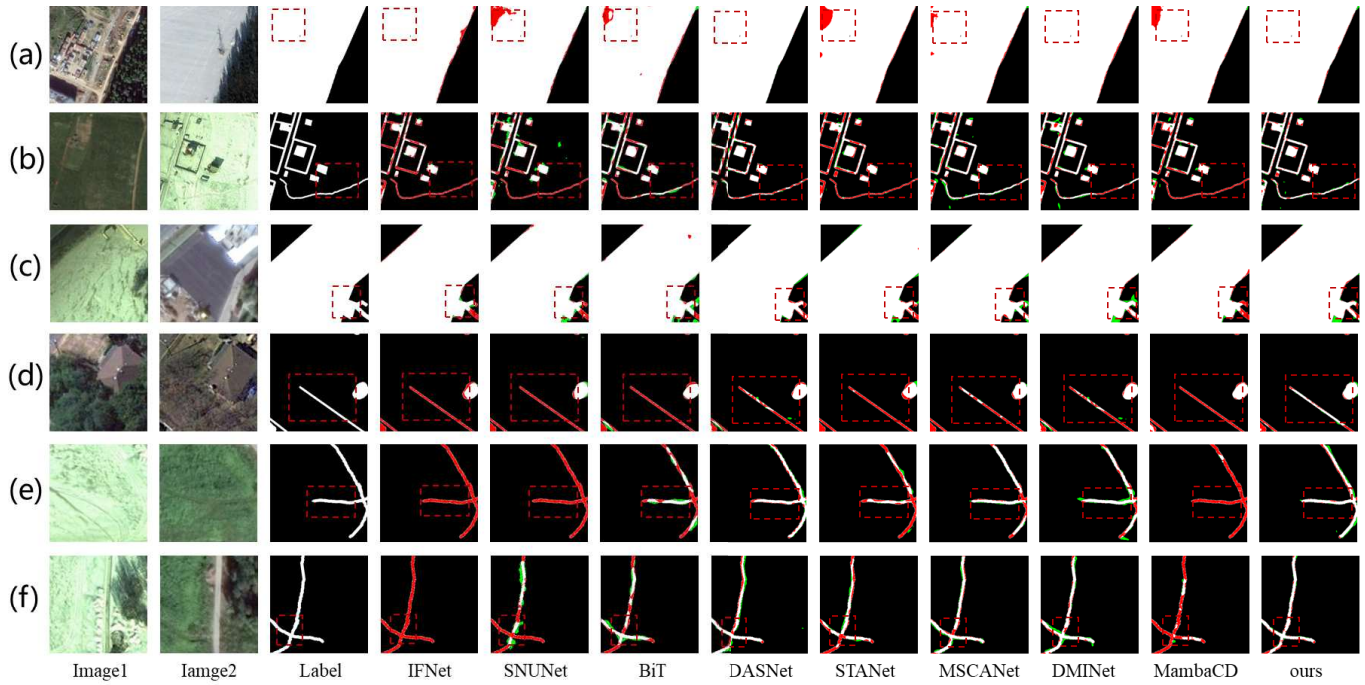


Fig. 5. Visual results of different models on test samples of CDD. (a)–(f) Six representative samples.

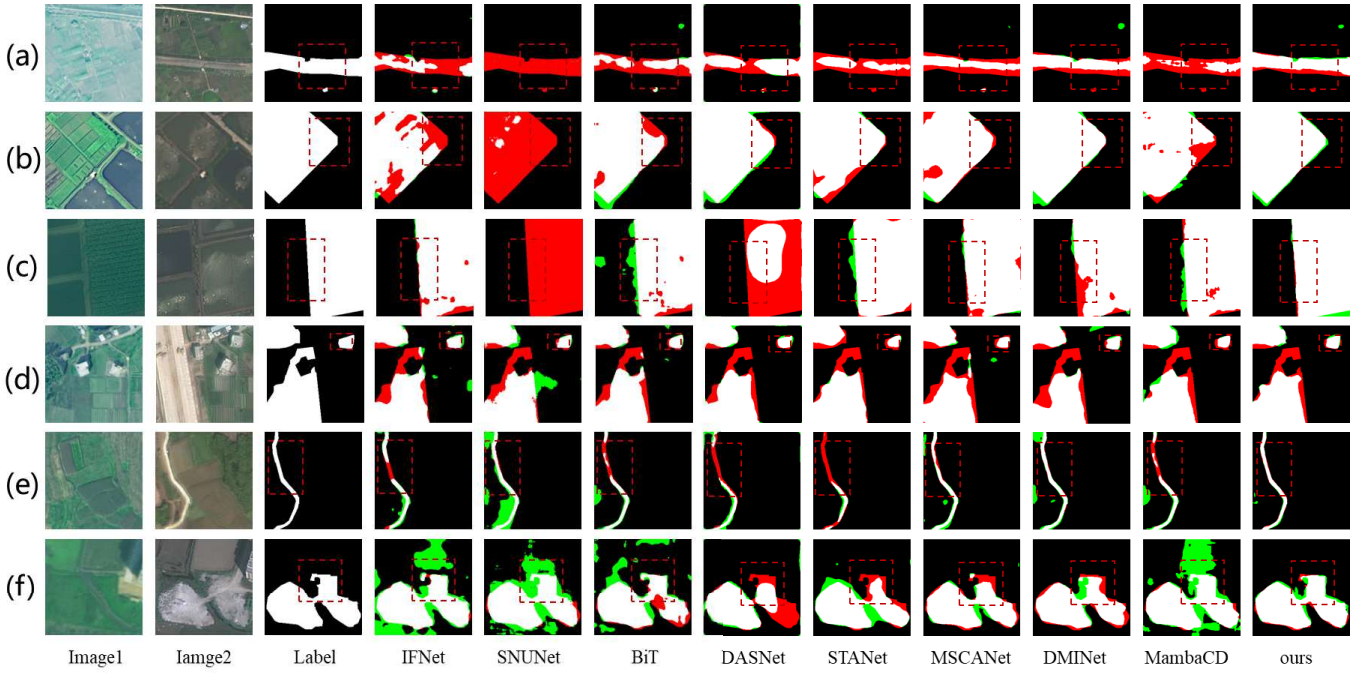


Fig. 6. Visual results of different models on test samples of CLCD. (a)–(f) Six representative samples.

ent climatic conditions. It is evident that in extremely cold weather, heavy snow covers the original roads. Furthermore, the small spectral differences interfere with the model's ability to effectively discriminate feature information. We utilize DFEM to enhance local feature details and integrate it with global context information, thereby enabling more effective identification of changed areas under external influences. As exemplified in Fig. 5(e) and (f), other SOTA approaches exhibit substantial false negative in road feature identification, particularly under challenging environmental conditions. In contrast, our DEIF-Mamba framework, which systematically integrates the DFEM and ISIF, demonstrates exceptional capability in precisely delineating change-target boundaries.

2) Visualization on CLCD: The CLCD primarily focuses on changes in farmland and bare land. Fig. 6(a), (e), and (f) illustrates the transition from original farmland to bare land or roads. When detecting the transition of vegetation or farmland to bare soil, other SOTAs often exhibit significant missed or

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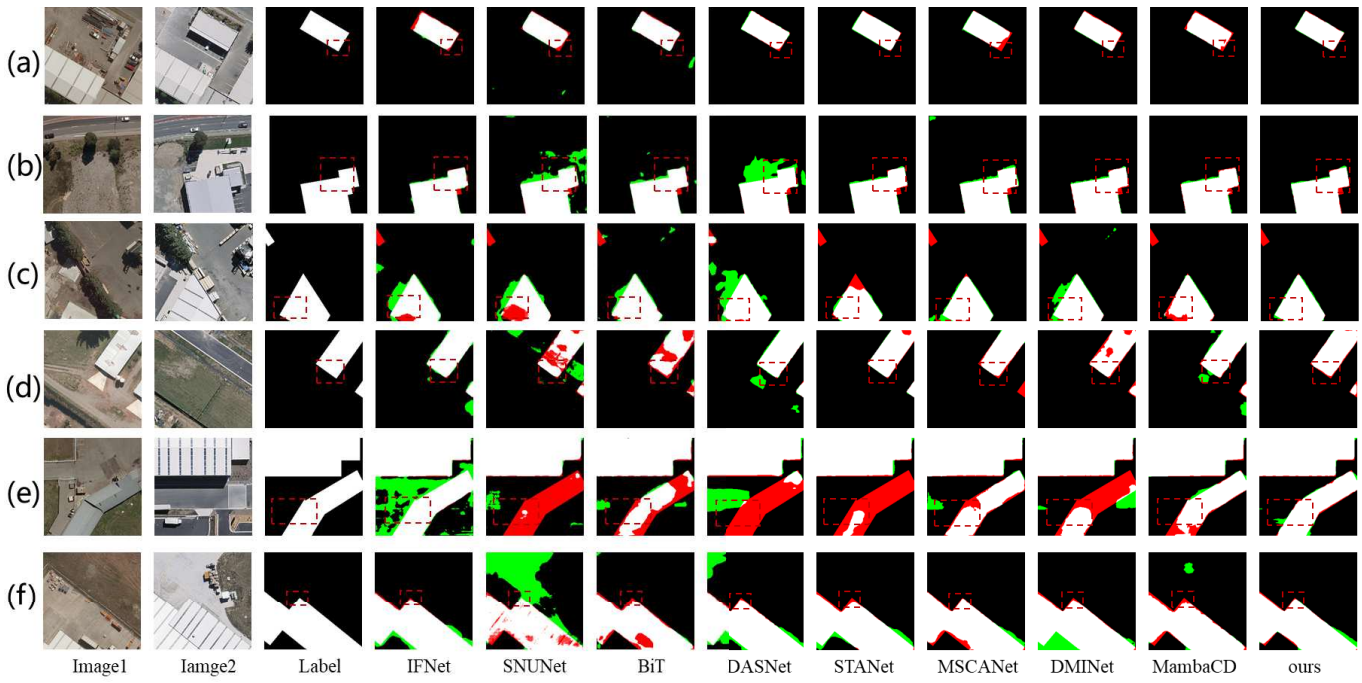


Fig. 7. Visual results of different models on test samples of WHU-CD. (a)–(f) Six representative samples.

TABLE III

ABLATION RESULTS (%) ON CD DATASET. THE SYMBOL "×" INDICATES THAT THE CORRESPONDING MODULE WAS REMOVED OR THE OPERATION WAS NOT PERFORMED. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	DFEM	ISIF	Multiscale	LEVIR-CD		CDD		WHU-CD		CLCD	
				F1(%)	IOU(%)	F1(%)	IOU(%)	F1(%)	IOU(%)	F1(%)	IOU(%)
DEIF-Mamba	×	×	√	80.41	67.24	94.65	89.84	87.76	78.19	73.20	57.22
	×	√	√	81.09	68.19	95.58	91.54	90.05	81.90	73.90	58.61
	√	×	√	80.92	68.01	94.86	90.55	89.63	81.19	74.76	59.43
	√	√	×	80.54	67.42	90.39	82.47	82.26	69.87	71.66	55.84
	√	√	√	**81.70**	**68.82**	**89.87**	**80.56**	**90.50**	**82.53**	**74.68**	**57.22**

false detections due to the lack of sufficiently rich discriminative feature information. We enhance feature representation by integrating the Gaussian filter with the NLB designed in DFEM and aggregating interactive feature information through ISIF, which enables the model to accurately identify change areas with maximum effectiveness. As exemplified in the farmland and rural trails shown in Fig. 6(b) and (e), DEIF-Mamba demonstrates superior boundary detection capabilities through the synergistic integration of DFEM and ISIF.

3) Visualization on Whu-CD: Fig. 7(a)–(f) reflects the typical demolition and reconstruction of buildings in the urbanization process. Due to the complexity of building boundaries and the similarity in color features, many SOTAs struggle to effectively distinguish feature changes. For instance, there are similar spectral curves at the junction between the building and the ground in Fig. 7(e) and (f), methods like MambaBCD may misclassify floors with similar color attributes to buildings as architectural changes. The visualization results indicate that DEIF-Mamba effectively leverages multiscale temporal feature information to accurately identify changes with weak contrast. Combined with the results in Table II, it is evident that DEIF-Mamba demonstrates superior detection performance.

D. Ablation Analysis

To assess the effectiveness of the proposed modules, we performed ablation experiments on four datasets, with the corresponding results summarized in Tables III and IV.

1) Differential Feature Enhancement Module: We evaluate the impact of removing DFEM on the overall model

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TABLE IV

COMPARISON RESULTS (%) OF DIFFERENT FUSION METHODS ON CD DATASET. THE SYMBOL "×" INDICATES THAT THE CORRESPONDING MODULE WAS REMOVED OR THE OPERATION WAS NOT PERFORMED. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

	ISIF	Element-wise	concatenation	LEVIR-CD		CDD		WHU-CD		CLCD	
				F1(%)	IOU(%)	F1(%)	IOU(%)	F1(%)	IOU(%)	F1(%)	IOU(%)
Method	×	√	×	81.20	68.35	95.44	91.29	88.76	79.80	74.54	59.20
	×	×	√	79.93	67.45	95.05	91.06	87.58	79.33	73.41	58.46
	√	×	×	**81.70**	**68.82**	**89.87**	**80.56**	**90.50**	**82.53**	**74.68**	**57.22**

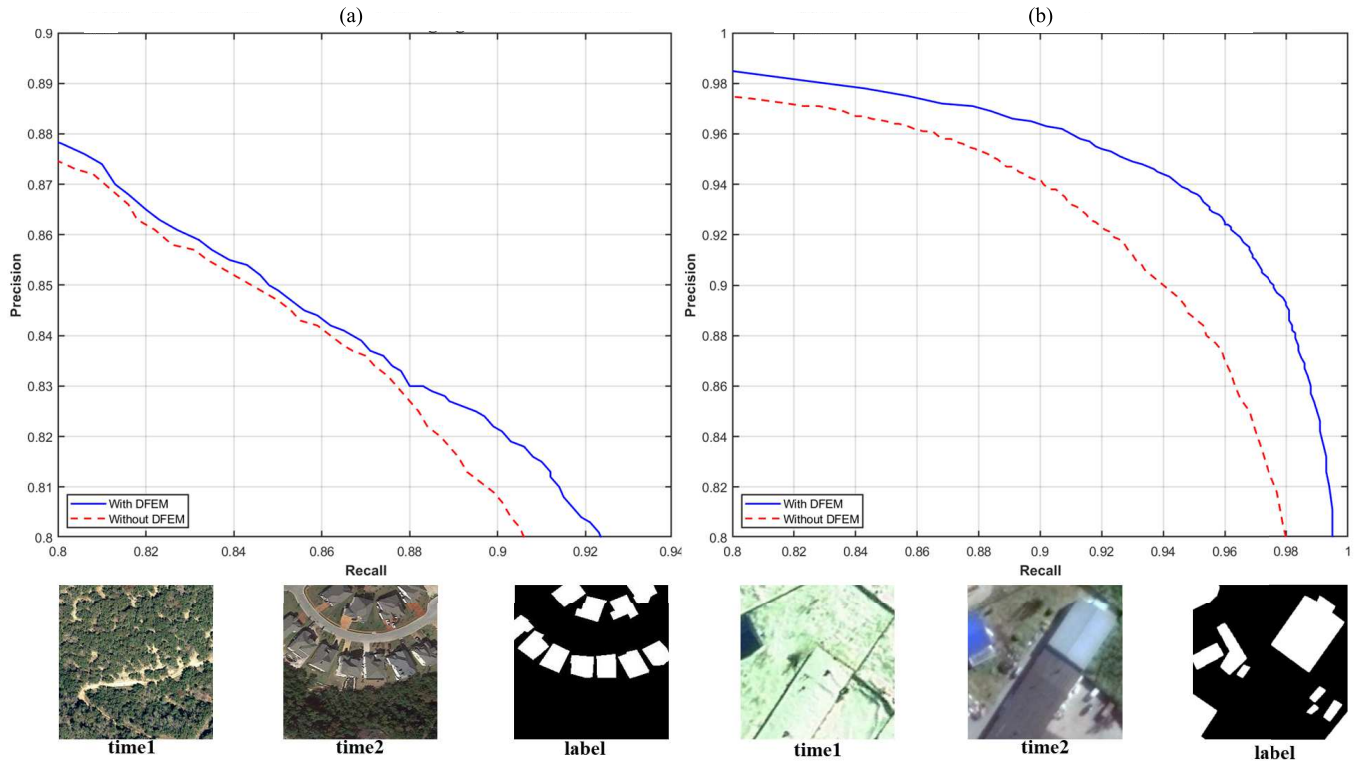


Fig. 8. Precision–recall curves on challenging cases in (a) LEVIR-CD and (b) CDD with and without DFEM.

performance, as presented in Table III. Compared with the proposed model, the F1 and IOU on the four datasets show a slight decrease after removing the DFEM. Specifically, on the CLCD, the F1 decreases by 2.25% and 2.88% in IOU. On the WHU-CD, the F1 reduced by 0.27% and 0.45% in IOU. For changes in vegetation, farmland, and densely packed buildings, the proposed DFEM effectively captures fine-grained feature details and integrates local and global contextual information, thereby enhancing the ability of DEIF-Mamba to accurately identify change characteristics. To further evaluate the contribution of the DFEM, we analyze precision–recall ($P - -R$) curves on challenging cases such as snow-covered regions in the CDD dataset and building shadows in the LEVIR-CD dataset. As illustrated by Fig. 8, incorporating DFEM leads to noticeably higher precision and recall for these complex scenarios, confirming its effectiveness in enhancing the model's discriminative power in difficult CD conditions.

2) Interscale Interactive Fusion: When ISIF is removed, the results are shown in Table III. The F1 on four datasets of LEVIR-CD, CDD, WHU-CD, and CLCD are 80.92%, 94.86%, 89.63%, and 74.76%, respectively. Quantitative indicators have shown that the fusion of features at different scales effectively combines spectrum and texture features, which aids in distinguishing dynamic changes from the static background. However, the simple linear combination fails to fully leverage the time-series information and is prone to causing feature information redundancy. The ISIF we proposed decomposes and interactively fuses scale features separately, effectively aggregating the discriminative information between features at different scales. This enhances the sensitivity of DEIF-Mamba to changed areas while minimizing feature redundancy. To

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TABLE V
EFFICIENCY COMPARISON ANALYSIS WITH DIFFERENT INPUT SIZES

Method	Average F1(%)	Flops(G)	Params (M)
IFNet	81.10	82.26	35.73
SNUNet	78.97	54.83	12.04
BiT	82.48	8.75	3.49
DASNet	80.66	107.98	55.31
STANet	81.37	18.07	16.93
MSCANet	84.54	14.80	16.42
DMINet	84.61	14.55	6.24
MambaBCD	84.00	44.83	85.52
DEIF-Mamba	85.57	13.42	22.66

fully verify the performance of the ISIF we proposed, we implemented alternative fusion strategies, including elementwise addition and feature concatenation, for comparison. As shown in Table IV, the ISIF strategy outperforms these baseline methods across multiple quantitative metrics, demonstrating its superiority in enhancing feature representation and model performance.

3) Multiscale Features in DEIF-Mamba: Features at various levels represent a progressive abstraction from low-level details to high-level semantic information. The integration of multiscale features enables the model to better accommodate different types of changes and diverse scenarios. As presented in Table III, DEIF-Mamba retains features from only a single scale, with F1 of 80.54%, 90.39%, 82.26%, and 71.66% on the LEVIR-CD, CDD, WHU-CD, and CLCD datasets, respectively. Local changes, such as small building renovations, are typically represented by low-level details, which facilitate the identification of fine-grained changes. In contrast, large-scale changes, such as urban expansion, are captured by high-level features, which provide strong global context awareness. We believe that combining multiscale feature information helps to mitigate the risk of excessive bias or neglect changed features at any particular scale, thereby enabling the model to effectively handle complex change scenarios.

E. Model Efficiency Assessment

To further assess the computational and storage efficiency of different models, we measure their performance using FLOPs and Parameters (Params). As shown in Table V, we have significantly enhanced computational efficiency compared with the MambaBCD. This improvement is attributed to our adjustment of the depth of the extracted multiscale features and the adoption of the ISIF strategy, which helps to mitigate information redundancy to some extent. In addition, we are able to maintain the highest average F1 across the four datasets, clearly demonstrating the effectiveness of our proposed DEIF-Mamba. Compared with other SOTAs, we found that when the input image size is 256×256 pixels, BIT has only 3.49 M Params and 8.75G computational complexity. BIT utilizes shared convolution weights and outputs a fixed number of channels during the feature extraction stage, which significantly reduces both the number of parameters and computational overhead by lowering the feature dimension. However, BIT employs simple feature stacking along channels, which limits its ability to capture interaction information effectively. Our proposed DEIF-Mamba achieves an average F1 that is 3.09% higher than BIT. Therefore, we believe that DEIF-Mamba not only achieves the highest F1 but also maintains high operational efficiency, thereby confirming its effectiveness.

TABLE VI
QUANTITATIVE ANALYSIS OF ACTIVATION INTENSITY IN TP AND FP REGIONS

Region	Method	CDD	CLCD	LEVIR	WHU-CD
TP	Baseline	93.92%	68.38%	68.55%	57.92%
	DFEM-Mamba	95.83%	77.62%	74.46%	62.30%
FP	Baseline	1.62%	3.86%	0.76%	0.49%
	DFEM-Mamba	1.48%	2.79%	0.57%	0.40%
TP/FP	Baseline	57.97	17.79	90.20	118.20
	DFEM-Mamba	64.75	27.82	130.63	155.75

F. Neural Network Visualization

To further explain the effectiveness of our proposed modules, we used MambaBCD as the baseline method and conducted a heatmap visualization experiment on the feature maps produced by the network at each stage, as shown in Fig. 9. We selected four pairs of representative images from four datasets and compared the outputs at various stages: after feature extraction, after the DFEM, after decoding, and the final results with the ISIF strategy applied. To compare the benchmark methods more intuitively, we modified the original Mamba encoder and uniformly used ResNet-18 as the feature extractor. The heatmap illustrates that after DFEM is applied, the model increasingly focuses on local features, such as roads or buildings. This demonstrates that the DFEM effectively captures local detail information while integrating global context, helping to reduce interference from nonchanging objects in complex scenes. Next, we compare the results obtained using the simple linear combination versus the ISIF strategy after the Mamba decoding layer. It is clear that by decomposing and interactively fusing features at different scales, the model improves the ability to utilize temporal change information and more effectively distinguish dynamic changes. By quantitatively calculating the activation intensity of the TP and FP regions through the model, the contribution of different branches to the change can be intuitively reflected, as shown in Table VI. The activation intensities of the TP and FP regions are represented by calculating the ratio of the positive example region pixels in the prediction result to the positive example region pixels in the label result, as well as

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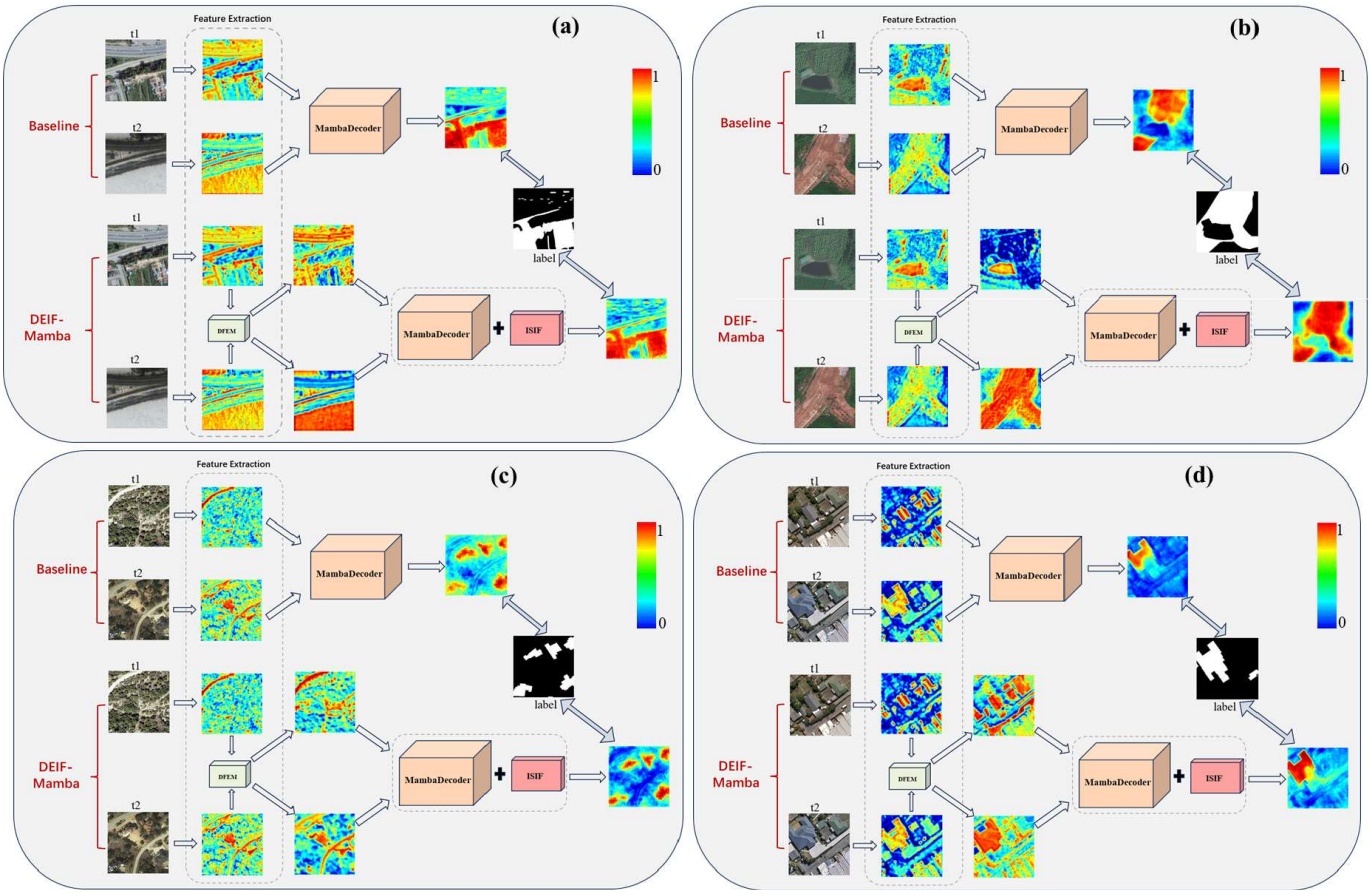


Fig. 9. Network visualization taking images from (a) CDD, (b) CLCD, (c) LEVIR-CD, and (d) WHU-CD as examples.

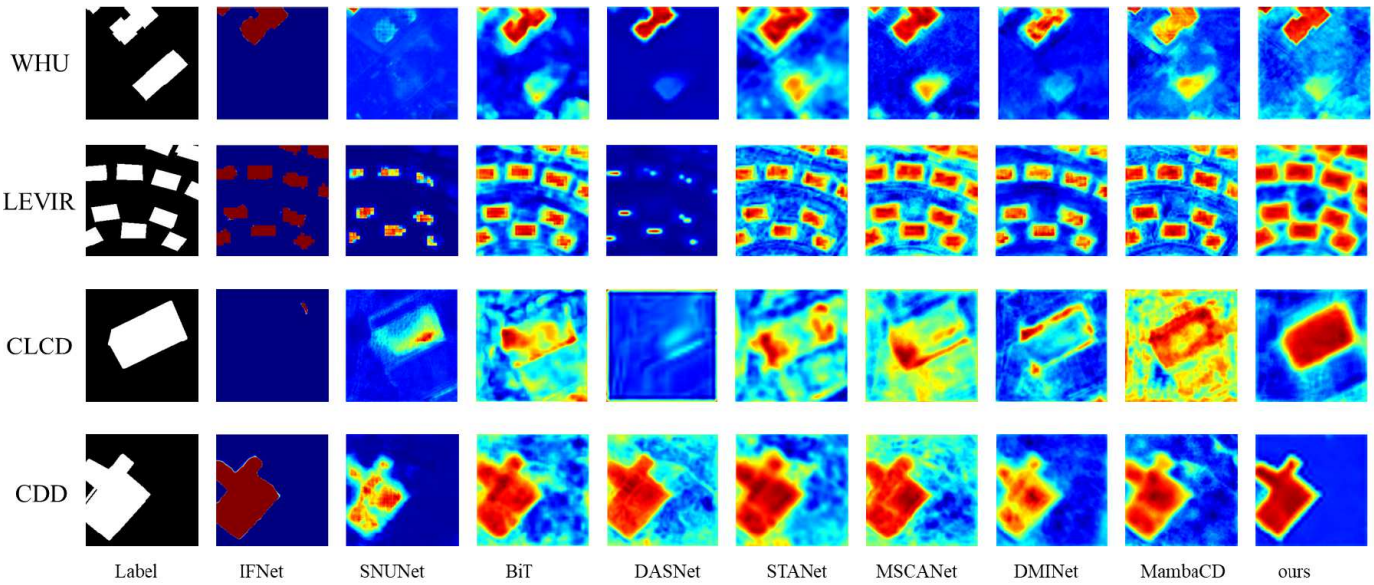


Fig. 10. Network visualization results of different models taking images from LEVIR-CD, CDD, CLCD, and WHU-CD as examples.

the ratio of the positive example region pixels in the prediction result to the negative example region pixels in the label result. We have selected one case pair from each of the four datasets (corresponding to Fig. 9) to calculate the activation intensity in TP and FP regions, and quantitative indicators demonstrate a significant improvement of activation intensity in the TP region. These results confirm that our proposed DFEM and ISIF effectively capture global–local change features while suppressing background noise interference in ground objects.

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In addition, we performed the heatmap visualization of the final output results from DEIF-Mamba and other SOTAs, as shown in Fig. 10. The results show that DEIF-Mamba effectively focuses on the true change areas, minimizing both false and missed detections, whether in densely built environments or areas with highly disruptive vegetation and farmland changes.

V. Conclusion

In the field of CV, existing visual Mambas traverse images from different directions through a unique scanning mechanism, enabling efficient global relationship modeling and enhanced performance. However, for RS image CD, they often overlook the critical role of local detail information in dense and complex scenes, resulting in certain interference in detecting scale changes or subtle changes. This article uses CNN-Mamba as the core network architecture and introduces two innovative parts, which are named as DFEM and ISIF. First, we extract shallow and deep abstract features of DT images at multiple scales through the ResNet backbone network. Next, the deepest DT features extracted are passed through the proposed DFEM to enhance local detail information. We then input DT features at different scales into the decoding layer of the Mamba architecture to learn global spatial context information. Finally, we apply the ISIF strategy to decompose and interactively fuse features at different scales, improving the discrimination of spatial change features and reducing information redundancy. Extensive experiments on four datasets demonstrate that our proposed DEIF-Mamba outperforms other SOTAs in terms of robustness. In particular, through heatmap visualization comparison, DEIF-Mamba demonstrates a more refined ability to capture local details in complex scenes. In future work, we will investigate the feasibility of using Mamba for CD in heterogeneous RS images.

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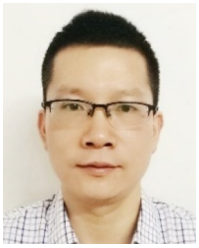
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